PRODUCT USER MANUAL

OMI_Baltic_TEMPSAL
OMI_CLIMATE_SST_BAL_area_averaged_anomalies
OMI_CLIMATE_SST_BAL_trend

Issue 4.0

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Record Table

Issue	Date	§	Description of Change	Author	Checked by	Validated by
1.0	2018/09/10	all	Creation of the document			C. Derval
2.0	2019/07/09	all	Update for the OMI 2018	Jacob L. Høyer		C. Derval
2.1	2020/09/09	all	Update for the OMI 2019	Jacob L. Høyer		C. Derval
2.2	2021/09/10	all	Update for the OMI 2020	Wiebke Margitta Kolbe, Ioanna Karagali, Sotirios Skarpalezos, Jacob L. Høyer		C. Derval
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3.1	2023/06/14	all	Update for the OMI 2022	Ioanna Karagali, Wiebke Margitta Kolbe, Jacob L. Høyer		Copernicus Marine Product Management
4.0	2024/05/31		Update for 2023, Change of climatology to 1991-2020, backwards extension to 1982	Ioanna Karagali, Wiebke Margitta Kolbe, Jacob L. Høyer		Copernicus Marine Product Management

GLOSSARY AND ABBREVIATIONS

MOI	Mercator Ocean International
NetCDF	Network Common Data Form
OMI	Ocean Monitoring Indicator
PUM	Product User Manual
QUID	Quality Information Document

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DATA ACCESS

After registration, you will be able to download our data. To assist you, our HelpCenter is available, and more specifically its section about download.

Information on operational issues on products and services can be found on our User Notification Service. If you have any questions, please contact us.

I INTRODUCTION

I.1. Summary

This document is the user manual for the CMEMS Ocean Monitoring Indicator (OMI) BALTIC_OMI_TEMPSAL. It contains 2 OMIs :

- OMI_CLIMATE_sst_BAL_area_averaged_anomalies
- OMI_CLIMATE_sst_BAL_trend

The OMIs provide estimates of monthly area averaged SST anomalies and trends for the Baltic Sea and North Sea, from 1982 to 2023. These estimates are derived from the Multi-Year Reanalysis product SST_BAL_SST_L4_REP_OBSERVATIONS_010_016. The time series of anomalies are computed with respect to a monthly climatology of observations from 1991-2020 (30 years). The climatology period is selected according to WMO recommendations (WMO, 2017) and the most recent practice from the U.S. National Oceanic and Atmospheric Administration practice (https://wmo.int/media/news/updated-30-year-reference-period-reflects-changing-climate).

The product is published on the Copernicus Marine dissemination server after automatic and delayed-mode quality controls. The product is available online and disseminated through the CMEMS Information System. Files downloaded are in NetCDF format and follow CF-1.7 convention.

Disclaimer: The quality of the product may vary during the proposed time series depending on the possible update of the system.

The two OMIs are described in the Quality Information Document (QUID) CMEMS-SST-OMI-QUID-010_027.pdf (see V).

I.2. History of changes

2018/09/10	1st publication of BALTIC_OMI_SST_area_averaged_anomalies_010_027 OMI product
2019/07	Temporal extension (+1 year)
2020/09	Temporal extension (+1 year)
2021/09	Temporal extension (+1 year)
2022/09	Temporal extension (+1 year)
2024/09	Temporal extension (+1 year), Backwards temporal extension (+11 years), New climatology 1991-2020 (30 years)

II DESCRIPTION OF THE PRODUCT

II.1. General information

Product(s) name	OMI_CLIMATE_sst_BAL_area_averaged_anomalies OMI_CLIMATE_sst_BAL_trend
Geographical coverage	Baltic Sea
Variables	Monthly Mean of Area Averaged Sea Surface Temperature Anomaly SST_ANOMALY Cumulative Sea Surface Temperature Anomaly Trend SST_TREND
Update frequency	Yearly
Available time series	From 1982 to year -1
Delivery mechanism	Marine Data Store
Temporal/Spatial resolution	yearly value, Baltic Sea mean value
Integration depth (for example..)	surface
Format	NetCDF CF1.7

II.2. Details of the datasets

OMI_CLIMATE_sst_BAL_area_averaged_anomalies contains the Monthly Mean of Area Averaged Sea Surface Temperature Anomaly
time [days since 1950-01-01 00:00:00] Reference time of SST field time **sst_amomaly [K]** Monthly Mean of Area Averaged Sea Surface Temperature Anomaly sea_surface_foundation_temperature
OMI_CLIMATE_sst_BAL_trend contains the Cumulative Sea Surface Temperature Anomaly Trend
sst_trend [K] Cumulative Sea Surface Temperature Anomaly Trend change_over_time_in_sea_water_temperature **Longitude [degrees_east]** longitude **Latitude [degrees_north]** latitude

II.3. Product System Description

This product provides the Baltic Sea SST TAC OMI, based on the products processed by DMI, covering the Baltic Sea and North Sea. The OMI time-series runs from Jan 1,1982 to December 31, 2023 and is constructed by calculating monthly averages from the daily level 4 SST analysis fields of the SST_BAL_SST_L4_REP_OBSERVATIONS_010_016 product. The times series of monthly anomalies have been used to calculate the trend in SST using Sen's method (Sen, 1968) with confidence intervals from the Mann-Kendall test (Kendall, 1975; Mann, 1945) (see V REFERENCES). The 30-year long climatology 1991-2020 is defined according to the WMO recommendation (WMO, 2017) and recent U.S. National Oceanic and Atmospheric

Administration practice (https://wmo.int/media/news/updated-30-year-reference-period-reflects-changing-climate).

III FILE FORMAT

The products are stored using the NetCDF format.

To know more about the NetCDF format, please follow this link:
What is the format of Copernicus Marine products ? NetCDF

To understand the differences between netCDF and Zarr, please consult this article:
how-to-choose-between-netcdf-and-zarr-format-using-the-toolbox

IV FILES NOMENCLATURE AND FORMAT

Information about nomenclature of files when downloaded can be found in this articles " How is defined the nomenclature of Copernicus Marine data? | Copernicus Marine Help Center "

IV.1. Nomenclature of files

OMI_CLIMATE_SST_BAL_area_averaged_anomalies files nomenclature when downloaded through the CMEMS Marine Data Store Services is based on the following nomenclature:

<datasetname>_<timestep>_P<productiontime>.nc

Where :

- **<datasetname>** is the dataset reference **(cmems_omi_climate_sst_bal_area_averaged_anomalies** or cmems**_omi_climate_sst_bal_trend).**
- **<timestep>** is the latest time step contained in the file**.**
- **<productiontime>** is the production time when the file was generated.
- **.nc** is the standard NetCDF filename extension.

Example: **cmems**_omi_climate_sst_bal_area_averaged_anomalies_2021_P20220701.nc

V REFERENCES

See the Quality Information Document (QUID): CMEMS-OMI-QUID-BALTIC-SST.pdf
https://catalogue.marine.copernicus.eu/documents/QUID/CMEMS-OMI-QUID-BALTIC-SST.pdf

(1) Kendall MG. 1975. Multivariate analysis. London: CharlesGriffin & Co; p. 210, 4.

(2) Mann HB. 1945. Nonparametric tests against trend.Econometrica. 13:245–259. p. 42

(3) Sen PK. 1968. Estimates of the regression coefficient based on Kendall's tau. J AmStatist Assoc. 63:1379–1389.

(4) WMO, Guidelines on the Calculation of Climate Normals, 2017, WMO-No-.1203